

★ WEEK 7 · PART 2 · PRACTICE

Testing for Cointegration, by Hand

The Engle-Granger Two-Step, and the Johansen Rank

From a residual test to a system verdict

Part 2's tests, with the arithmetic shown. We run the Engle-Granger two-step - regress, then test the residual against its special critical value - and then use the Johansen trace test to step down and count how many equilibria hold a pair together.

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PART 2 · APPLICATIONS

Prove It Before You Trade It

Case A: the Engle-Granger two-step - regress one price on the other, then test the residual for stationarity against the special critical value. Case B: the Johansen trace test, stepping down to find the cointegration rank. Then a Claude Code lab.

What Part 2 Told Us

Now we test it - with numbers

1

First confirm each series is $I(1)$

Cointegration is a statement about non-stationary series. Both prices must be $I(1)$ before any cointegration test makes sense.

2

Engle-Granger tests the residual

Regress one price on the other, then test the residual for stationarity - against special critical values, because the residual is estimated.

3

Johansen counts the equilibria

A symmetric system test finds the cointegration rank - how many independent stationary combinations hold the series together.

Everything here is a calculator exercise: one residual test, one trace-statistic step-down.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from two $I(1)$ price series.
- Run the residual test, or the trace test.
- Then read the verdict.

What you should watch

- Why the residual needs a special critical value.
- How the trace test steps down through ranks.
- How both reach the same conclusion.

The Engle-Granger Two-Step

Regress, then test the residual

Engle-Granger is the quick check for a single pair. Step one regresses one price on the other and keeps the residual - the spread. Step two runs an ADF test on that residual. Because the residual was estimated, we compare its statistic to a special, more demanding critical value. We work both steps and reach a verdict

The setup

- Two $I(1)$ prices, Y and X.
- 180 observations.
- A candidate pairs trade.

What we'll do by hand

- Regress Y on X for the spread.
- Run ADF on the residual.
- Compare to the Engle-Granger cutoff.

The Two-Step Procedure

A regression, then a residual test

Step 1: $Y_t = \alpha + \beta X_t + \hat{s}_t$ $\longrightarrow$ Step 2: ADF on $\hat{s}_t$ ($\Delta \hat{s}_t = \rho \hat{s}_{t-1} + \varepsilon_t$)

Step one

- Regress the level of Y on the level of X.
- The residual is the estimated spread.
- Beta is the cointegrating vector.

Step two

- Test that residual for a unit root.
- A stationary residual means cointegration.
- Use the special critical value.

Regress, Then Test

The spread and its ADF statistic

$$\text{Step 1: } Y_t = 3.65 + \mathbf{1.98} X_t + \hat{s}_t$$

$$\text{Step 2: ADF } t\text{-stat on } \hat{s}_t = \mathbf{-5.07}$$

The regression gives a hedge ratio of 1.98 and a residual spread. Running an ADF test on that spread yields a t-statistic of -5.07 - strongly negative, hinting the spread reverts. But we must check it against the right cutoff.

The Verdict

Against the special critical value

$$-5.07 < \underbrace{-3.34}_{\text{Engle-Granger 5\%}} < \underbrace{-2.86}_{\text{plain DF 5\%}} \Rightarrow \text{reject unit root} \Rightarrow \text{cointegrated}$$

Why a special value

- The residual was estimated, not observed.
- That makes stationarity easier to find by chance.
- So the cutoff is more negative than plain DF.

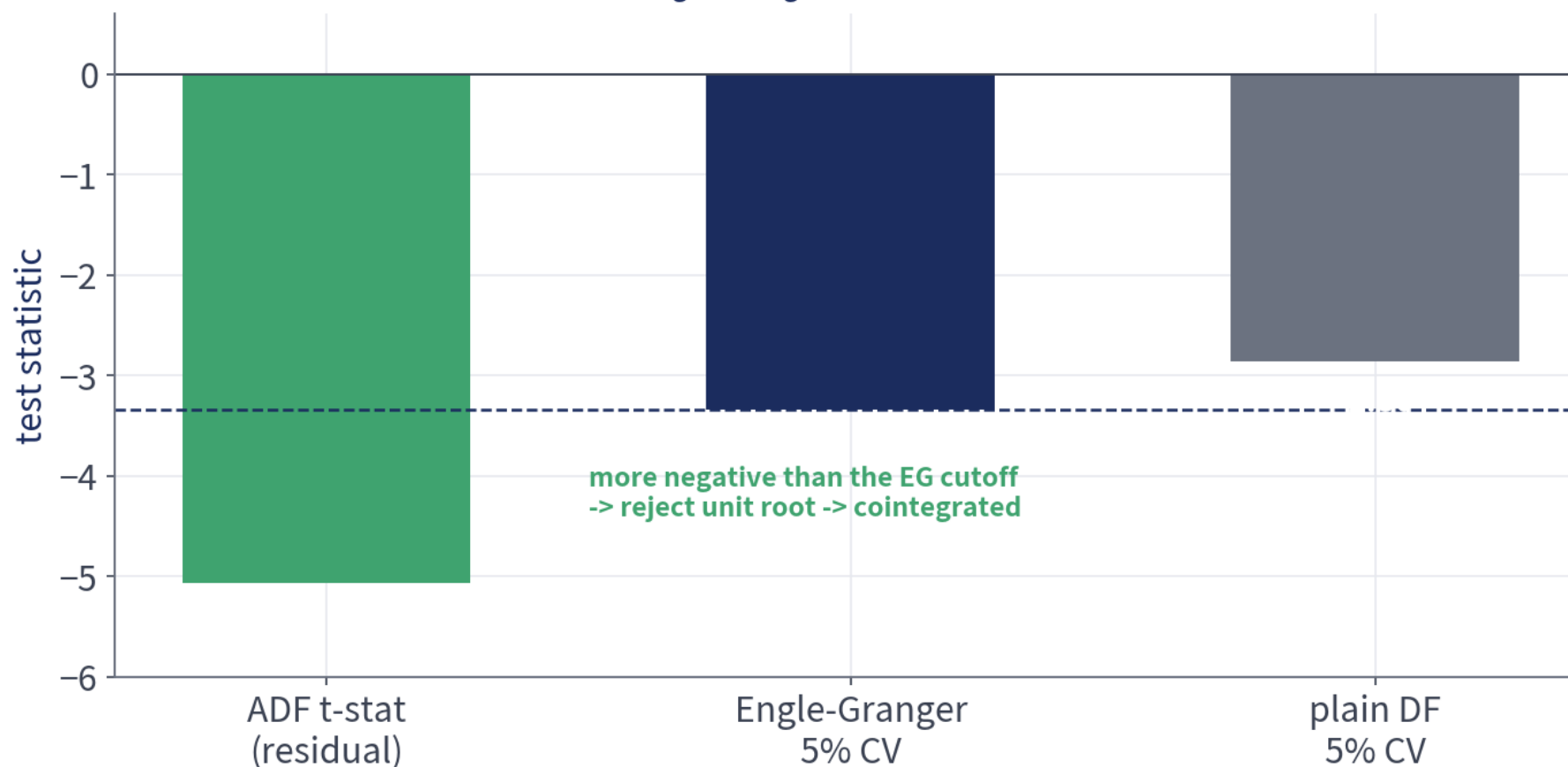
The decision

- At -5.07, the statistic clears -3.34.
- Reject the unit root - the spread is stationary.
- The pair is cointegrated.

Clearing the Right Bar

The residual's ADF statistic of -5.07 sits well below both cutoffs. Note that the Engle-Granger critical value (-3.34) is more demanding than plain Dickey-Fuller (-2.86) - using the wrong, easier bar would over-declare cointegration. Here the statistic clears even the tougher bar

Case A: the residual's ADF stat clears the Engle-Granger critical value



Simple, but Watch the Details

Two subtleties that matter

The special critical value

- Never use plain DF cutoffs on a residual.
- The estimated residual biases the test.
- Engle-Granger tables fix this.

The asymmetry

- Regressing X on Y gives a different beta (0.50).
- And a slightly different statistic (-5.04).
- Near the boundary, the two can disagree.

The Johansen Rank

Count the equilibria, symmetrically

Engle-Granger picks a dependent variable, so it isn't symmetric. Johansen treats all series alike and tests the whole system at once, returning the cointegration rank - the number of independent stationary combinations. We use the trace statistic and step down: reject rank zero, then stop when we can no longer reject

The rank r

- $r = 0$: no cointegration.
- $r = 1$: one equilibrium - a tradable pair.
- $r = 2$: both series already stationary.

What we'll do by hand

- Compute the trace statistic at $r = 0$.
- Then at r at most 1.
- Stop where we cannot reject.

The Trace Statistic

Step down until you cannot reject

$$\text{trace}(r) = -T \sum_{i=r+1}^k \ln(1 - \hat{\lambda}_i) \quad (\text{test } r=0, \text{ then } r \leq 1, \text{ step down until you cannot reject})$$

The procedure

- Start at rank zero - the null of no equilibrium.
- If the trace exceeds its critical value, reject.
- Move up a rank and test again.

The stopping rule

- Keep going until you cannot reject.
- That rank is the answer.
- It counts the cointegrating vectors.

Test Rank Zero

Is there any cointegration at all?

null hypothesis	trace statistic	5% critical value	decision
$r = 0$ (none)	28.6	15.49	REJECT

At rank zero, the trace statistic of 28.6 far exceeds the critical value of 15.49, so we reject the null of no cointegration. There is at least one equilibrium holding the pair together - now we ask whether there might be a second.

Trace 28.6 beats the 15.49 bar: reject no-cointegration, at least one equilibrium exists.

Step Up to Rank One

Where the step-down stops

null hypothesis	trace statistic	5% critical value	decision
$r \leq 1$ (one)	1.4	3.84	cannot reject

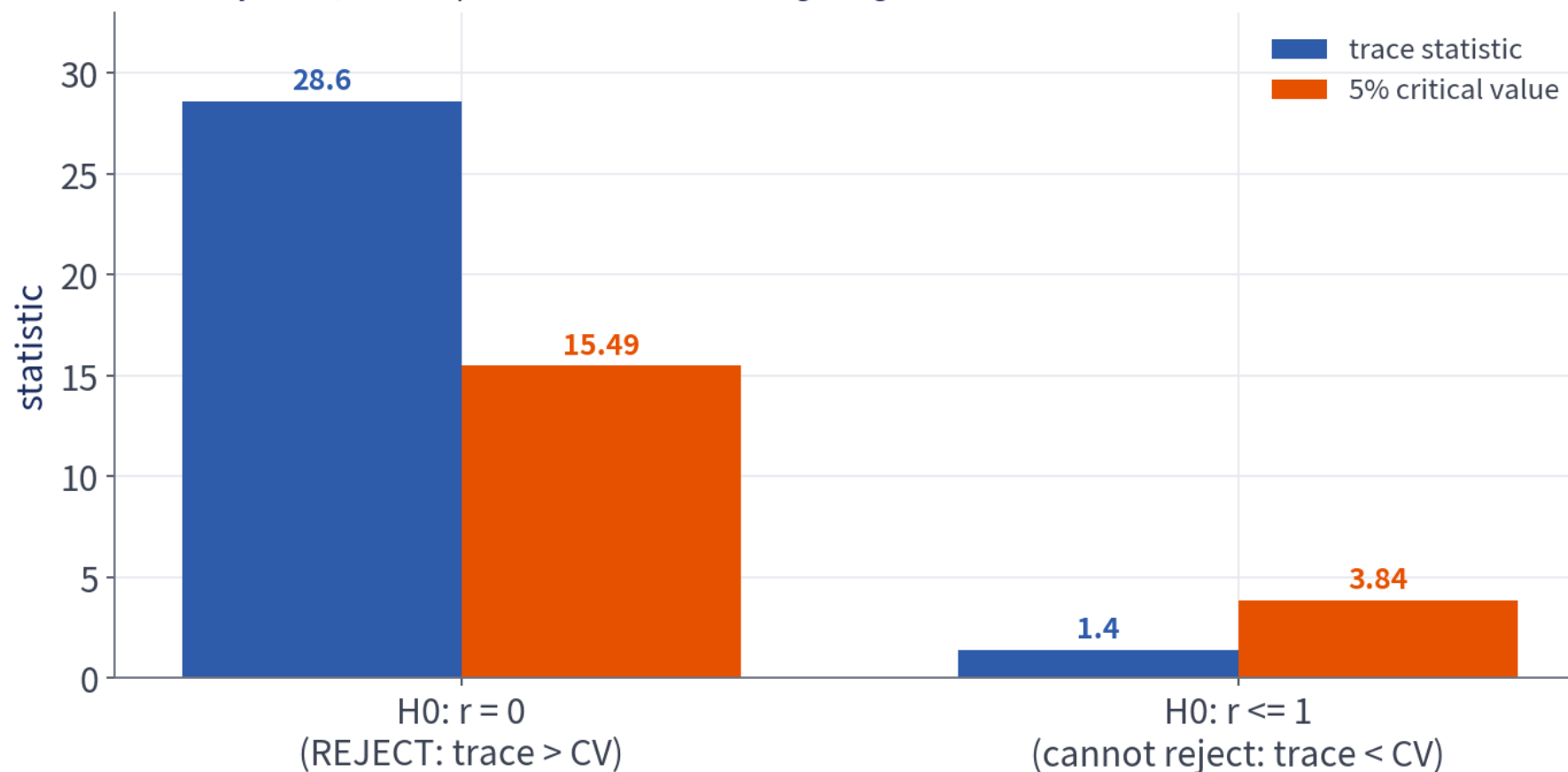
Now the trace statistic of 1.4 sits below the critical value of 3.84, so we cannot reject rank at most one. The step-down stops here: the rank is exactly one - a single cointegrating vector holds the pair together.

Fail to reject $r \leq 1$: the rank is exactly 1, so the pair is cointegrated.

Step Down to the Rank

At rank zero the trace statistic towers over its critical value, so we reject - there is cointegration. At rank at most one, the statistic drops below the bar, so we stop. The step-down lands on rank one: exactly one equilibrium - the symmetric confirmation of Case A's verdict

Case B: reject $r=0$, then stop at $r \leq 1 \rightarrow$ rank = 1 cointegrating vector



Why Reach for Johansen

When the system matters

Its advantages

- Symmetric - no dependent variable to pick.
- Handles many series at once.
- Counts multiple equilibria via the rank.

When it earns its keep

- Baskets of three or more assets.
- When Engle-Granger's asymmetry bites.
- When you need the number of equilibria.

Two Tests, One Verdict

Quick check on system test

Both routes ask the same question - is there a stationary combination? - and here both say yes. Engle-Granger is the fast two-step for a single pair, testing an estimated residual against a special cutoff. Johansen is the symmetric system test that counts equilibria via the rank. Use Engle-Granger to screen a pair, Johansen to

Engle-Granger (Case A)

- Regress, then ADF the residual.
- Statistic -5.07 clears -3.34.
- Fast, but asymmetric.

a

Johansen (Case B)

- Trace test, step down the rank.
- Reject $r=0$, stop at $r \leq 1$.
- Symmetric, counts equilibria.

Reproduce Both Tests

Direct Claude Code to check your arithmetic

GOAL Run the Engle-Granger two-step and the Johansen trace test on the same pair.

[INPUT] Prompt · Claude Code

Write and run `part2_practice.py` that, on a cointegrated pair Y, X (build one, or reuse Part 1's):

CASE A - Engle-Granger:

- * regress Y on X , keep the residual;
- * run `adfuller` on the residual (`regression='n'`);
- * print the t-stat and compare to -3.34 .

CASE B - Johansen:

- * run `coint_johansen(det_order=0, k_ar_diff=1)`;
- * print the trace stats and 5% critical values;
- * report the rank.

Confirm both say cointegrated.

Check against the slides

- $\beta = 1.98$, ADF -5.07 .
- clears EG -3.34 .
- trace $r=0$: $28.6 > 15.49$.
- trace $r \leq 1$: $1.4 < 3.84 \rightarrow r=1$.

Deliverable: `part2_practice.py` whose printed numbers match the two-step and the trace test.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 Why does Engle-Granger need special critical values?

The residual is estimated, biasing the test

Q2 How does Johansen determine the rank?

Step down until you cannot reject

Q3 When would you prefer Johansen?

For systems of three or more, or by symmetry

All three come from the Engle-Granger two-step and the Johansen trace test.

Three Pitfalls to Avoid

Where cointegration testing trips people

Common mistakes

- "Use plain Dickey-Fuller cutoffs on the residual."
- "Y-on-X and X-on-Y must give the same answer."
- "Skip checking that each series is $I(1)$."

The reality

- Use Engle-Granger critical values - the residual is fitted.
- The regression is asymmetric; results can differ.
- Cointegration only makes sense for $I(1)$ series.

Two Tests to Remember

Part 2 applications in one screen

1

Engle-Granger: regress, then test the residual

Regressing Y on X gave a hedge ratio of 1.98; the residual's ADF statistic of -5.07 cleared the special Engle-Granger cutoff of -3.34 - more demanding than plain DF because the residual is estimated. The pair is cointegrated.

2

Johansen: step down to the rank

The trace statistic of 28.6 rejected rank zero (critical 15.49), then 1.4 could not reject rank at most one (critical 3.84). The step-down stopped at rank one - a single cointegrating vector, symmetrically confirming the same verdict.

3

Screen with one, analyze with the other

Engle-Granger is a fast two-step for a single pair; Johansen is the symmetric system test that counts equilibria. Both agreed here - the pair is genuinely cointegrated and ready for the error-correction model.

Next — Part 3 practice: the error-correction model and building the pairs trade itself.